

Renewable-Energy PoC — Executive Summary

Chilean renewable-energy data platform on Databricks | updated 2026-07-28

Overview

A proof of concept for a renewable-energy data platform on Databricks using a medallion (bronze → silver → gold) lakehouse. It ingests real data from independent public sources, turns it into analytics-ready metrics across three domains (Weather · Resource · Conglomerate), and is fully governed, monitored and dashboarded — all in a dev environment, on isolated branches; production is the project lead's.

Data sources connected — all on real data

Domain	Source	Access	Cadence
Weather	NASA POWER · DMC · MinEnergía/DGF	open / token / session	historical hourly + real-time + world daily
Resource (generation)	CEN / Coordinador	public by law (Art. 72-8, Ley 20.936) — no cost	daily per plant
Conglomerate (country)	Our World in Data (OWID)	open CSV	annual

Key unblocks:

- CEN generation was assumed to need a paid/registered SIPUB account. It is public information by law and reachable with the public key the Coordinador's own site exposes — no registration, no cost.
- Historical weather with the same fields as the DMC station network is served cleanly by NASA POWER (10-year hourly full load, 876,720 rows), and extended to a 12-city world benchmark (all continents) for Chile-vs-world comparison.

Weather officialized in a dedicated schema

All weather is consolidated in three purpose-built schemas — weather_bronze / weather_silver / weather_gold (dev catalog). Each layer holds per-source tables (nasa_power, dmc, dgf) plus one consolidated all_sources table carrying a source column, so every weather record — historical, real-time or modeled — lives under one governed roof (bronze 882k · silver 881k rows).

Results (verified in dev)

- Medallion runs end-to-end for all three domains (bronze → silver → gold), cleaned and self-documented (schema + table comments, symmetric naming).
- Weather: 10-year historical hourly loaded (physically validated — Atacama shows the highest irradiance), modeled as a star schema + marts (daily/monthly/KPI), consolidated across sources, and served through three AI/BI dashboards:
 - Chile & Rest of World (one dashboard, two tabs) — seasonality, trends, max/min/means for temperature, GHI, pressure and wind; Rest-of-World tab shows the Northern-vs-Southern hemisphere inversion across 12 cities.
 - Chile & Rest of World — 2025-2026 YTD — the same views on the latest data.
 - Weather NASA — time series + resource KPIs.
- Resource: real per-plant generation (coordinado / real / reducciones).
- Conglomerate: Chile-vs-World renewables lead grew +19 → +38 (2017-2024).

Governance & data quality (Databricks-native)

- Data contracts documented for every medallion step; DLT expectations enforce quality in-pipeline.
- Data glossary (business + technical) and Unity Catalog lineage + column/table comments.

- Observability: three Lakehouse Data-Quality Monitors on the gold tables (profiling + drift, auto-generated dashboards) + a SQL alert on failed expectations.

Status & next steps

- Delivered and verified in dev, consolidated in delivery ticket Trello #74. The five PRs (#24–#28) were closed without merge as the team's main advanced on a separate track; the work remains on the fork branches, ready to be re-integrated (CI fix is the priority — the old CI deploys to prod on approval and runs no tests).
- A cadence reality was documented: most public sources are not real-time (DGF/MinEnergía are historical; CEN lags months). The DMC feed is the only real-time measured source; NASA POWER covers history and the world benchmark.

Explore / test the live results

- Weather — Chile & Rest of World dashboard (2 tabs):
<https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18a95d5851bb1a33415a71141a908?o=7474645896934260>
- Weather — Chile & Rest of World, 2025–2026 YTD:
<https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18a9c8d2f19a48485521da3f5a4f6?o=7474645896934260>
- Weather NASA dashboard (series + KPIs):
<https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18771593d19568e23e322277007c3?o=7474645896934260>
- Data-quality monitor (Resource gold):
<https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f1809adb1a199384b80e96c469a206?o=7474645896934260>
- Pull requests: https://github.com/oxiboy/poc_databricks_evaluateserve/pulls